VISHWANATH M M

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PROFESSIONAL SUMMARY

AI and Machine Learning undergraduate with hands-on experience in software engineering, backend development, systems programming, automation testing, and open-source software. Proficient in Python, Rust, Selenium WebDriver, REST APIs, SQL, Git, and software development practices. Founder and Lead Developer of the Ferrite programming language and maintainer of multiple Python packages published on PyPI. Experienced in building reliable software systems, automated test scripts, backend applications, and developer tools, with strong analytical, problem-solving, and debugging skills.

EDUCATION

Bachelor of Engineering — Artificial Intelligence & Machine Learning 2023 – Present
Bapuji Institute of Engineering and Technology, Davangere **CGPA: 7.94 / 10.0**

Class 12 (Pre-University) 2023
Karnataka State Board **Percentage: 85.16%**

Class 10 (SSLC) 2021
Central Board of Secondary Education. **Percentage: 73.33%**

TECHNICAL SKILLS

Programming: Python, Rust, SQL.
Software Testing: Selenium WebDriver, Test Automation, Automated Test Scripts, API Testing.
Backend & APIs: REST APIs, REST API Design, Backend Development.
Databases: MongoDB, PostgreSQL
Systems Programming: Compiler Design, Systems Programming, Lexer/Parser Construction, CLI Tooling.
AI / ML: Machine Learning, Natural Language Processing (NLP), LLM Integration.
Web: HTML, CSS, React.
DevOps & Version Control: Git, GitHub, Docker, Linux, PyPI.
Professional Skills: Analytical Thinking, Problem Solving, Debugging, Communication, Collaboration

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EXPERIENCE

Founder & Lead Developer — Ferrite Programming Language **2025 – Present**
Website: ferrite-lang.org
GitHub: github.com/vishwanathdvgmm/ferrite

- **Architected and developed Ferrite**, a statically-typed, ahead-of-time compiled systems programming language written in Rust, optimized for machine learning workflows and tensor manipulation.
- **Built a complete 5-stage compiler pipeline** (Lexing, Parsing, Semantic Analysis, IR Generation, and Native Compilation), featuring a recursive descent parser and a multi-pass type-checker backed by **LLVM**.
- **Engineered a fully expression-oriented architecture** supporting deep scoping, advanced control flow (skip, stop, return), and native dynamic collections (List<T>, Map<K, V>) backed by scope-based RAII memory management.
- **Developed a rich developer ecosystem**, including an official **VS Code extension** (syntax highlighting, formatting), a comprehensive standard library, and an interactive browser-based JavaScript playground.
- **Established a robust CI/CD pipeline** via GitHub Actions to automate rigorous unit/integration testing (via Insta snapshots), static analysis (cargo clippy), and automated deployment of the documentation website.

PROJECTS

Open-Source Python Packages (PyPI) 2024 – 2026

Open-Source Software Projects

Tech Stack: *Python, PyPI, PEP 508, Packaging, Dependency Resolution, CLI Development, Automated Testing.*

- Developed and published **two open-source Python packages**, **Pyvenvmerge** and **Symjoy**, with modular architectures, automated testing, semantic versioning, and PyPI distribution.
- Built **Pyvenvmerge**, a CLI utility for safely merging Python virtual environments through dependency reconstruction, with **PEP 508 parsing, conflict resolution, dependency analysis, and semantic validation**.
- Added **dry-run diagnostics, JSON reporting, lockfile export, and deterministic rebuilding**, culminating in the stable **v1.0.0** release.
- Developed **Symjoy**, an open-source Python library with a **JSON-driven, metadata-based architecture** for Unicode symbols, emojis, arrows, mathematical operators, currency symbols, and miscellaneous characters.
- Built an **indexed search engine** with keyword, alias, and token-based retrieval, deterministic ranking, multi-token queries, **semantic grouping**, category APIs, automated testing, GitHub Actions CI, and PyPI distribution.
- **GitHub:** github.com/vishwanathdvgmm/pyvenvmerge | github.com/vishwanathdvgmm/symjoy

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Software Intelligence Platform (SIP) 2026 – Present

Research Project — Adaptive RAG & Software Intelligence

Tech Stack: *Python, RAG, Hybrid Search, BM25S, Qdrant, PostgreSQL, Cross-Encoder Reranking, FastAPI, Tauri, React, TypeScript.*

• Designing an adaptive RAG architecture that analyzes query intent and complexity to dynamically select retrieval strategies, combining semantic search, BM25S lexical retrieval, and version-aware metadata filtering.

• Researching a multi-stage retrieval pipeline using Reciprocal Rank Fusion (RRF), cross-encoder reranking, context optimization, evidence gating, and bounded retrieval retry/reformulation to improve retrieval precision and grounded generation.

• Architecting a software intelligence platform that builds version-aware software Experts from documentation, release notes, GitHub, community knowledge, and software-specific sources, with measurable evaluation against traditional RAG baselines.

Blockchain-Based AI-Enabled Digital Evaluation System 2025

Academic Project

Tech Stack: *Go, Python (FastAPI), PostgreSQL, React, Blockchain (BoltDB), IPFS, Google GenAI*

- **Architected** a microservices-based digital evaluation platform, orchestrating end-to-end grading workflows with a high-performance **Go** core and a **Python** AI microservice.
- **Engineered** an intelligent evaluation pipeline using **OpenCV**, **PyMuPDF**, and **Google GenAI** to automate script metadata extraction and validate grading accuracy.
- **Developed** a tamper-evident blockchain transaction chain using BoltDB (SHA-256 anchoring) alongside **PostgreSQL** for relational data, ensuring strict data integrity for academic records.
- **Integrated** the InterPlanetary File System (**IPFS**) to provide resilient, decentralized storage for uploaded evaluation PDF artifacts.
- **Built** a secure, Role-Based Access Control (RBAC) **React** web interface enabling 5 distinct user personas (Admins, Evaluators, Students, etc.) to securely submit, grade, and verify records.
- **GitHub:** github.com/vishwanathdvgmm/digital-eval-system

CERTIFICATIONS & TRAINING

- **Hashgraph Developer Course** — The Hashgraph Association 2025
- **Building AI-Powered Search with MongoDB Vector Search** — MongoDB 2026
- **Artificial Intelligence Training** — Acmegrade 2026